Problem Statement

AI-Driven Multi-Agent Regulatory Compliance System

Introduction & Motivation

The integration of Generative AI (GenAI) and Large Language Models (LLMs) has significantly transformed regulatory compliance by enabling automated policy analysis, intelligent gap detection, and proactive compliance monitoring. AI-powered solutions can assist in mapping regulations to policies, identifying compliance gaps, and preparing for audits.

Despite the critical importance of regulatory compliance, mapping internal policies to regulatory requirements remains a complex, manual process. Many organizations struggle to keep pace with regulatory changes and maintain comprehensive compliance documentation. ComplianceIQ is designed to bridge this gap by leveraging LLMs and multi-agent architectures to automate compliance assessment, gap detection, and remediation planning.

Business Problem Overview and Potential Impact

Challenges in Regulatory Compliance

- Multi-Jurisdiction Complexity – Organizations must comply with multiple regulatory frameworks (IRDAI, MAS, etc.).
- Policy-Regulation Mapping – Connecting internal policies to specific regulatory requirements is time-consuming.
- Gap Identification – Finding compliance gaps before audits requires comprehensive analysis.
- Documentation Burden – Preparing evidence and documentation for audits is resource-intensive.

Who Can Benefit from ComplianceIQ?

User	Use Case
Software Engineers (SDEs)	Build scalable AI-powered compliance analysis systems.
AI/ML Professionals	Develop multi-agent systems for regulatory document intelligence.
Compliance Officers	Automate gap detection and compliance assessment across regulations.
Risk Managers	Identify and prioritize compliance risks before they impact operations.

Internal Auditors	Streamline audit preparation with automated evidence collection.
Legal Teams	Track regulatory changes and assess policy impact.
Insurance Operations	Ensure policy alignment with IRDAI, MAS, and other requirements.

Potential Impact

Companies like Ascent, Clausematch, and RegTech solutions are already leveraging AI for compliance automation. ComplianceIQ aims to build upon this trend by providing multi-agent compliance analysis, intelligent gap detection, and knowledge graph-based regulation mapping that reduces compliance risk and audit preparation time.

Key Problem Statement

> *"Develop an intelligent regulatory compliance platform that analyzes company policies against regulatory requirements (IRDAI, MAS) to identify compliance gaps and provide remediation recommendations. The system will extract regulatory requirements, map them to internal policies, and assess compliance coverage. The solution will feature specialized agents for regulation monitoring, gap detection, and policy analysis, providing comprehensive compliance dashboards and audit-ready documentation."*

Core Objectives

- Implement Multi-Agent Compliance Analysis – Deploy specialized agents for regulation, gap, and policy analysis.
- Develop Semantic Regulation Understanding – Use embeddings to match regulations with relevant policies.
- Enable Automated Gap Detection – Identify compliance gaps with severity scoring and remediation priorities.
- Provide Knowledge Graph Insights – Map regulation-policy relationships across compliance domains.
- Ensure Audit-Ready Documentation – Generate evidence trails and compliance reports for regulators.

System Features

Data Processing & Analysis

- Load and process regulatory PDFs and policy documents
- Extract requirements, mandatory actions, and evidence needs
- Categorize documents by compliance domain

Multi-Agent Compliance Intelligence

- Regulation Monitor Agent: Analyze regulatory requirements, extract mandatory actions and evidence needs
- Gap Detector Agent: Compare regulations against policies, identify gaps with severity scoring
- Policy Analyzer Agent: Evaluate policy coverage across regulatory domains, provide coverage scores

Compliance Assessment & Scoring

- Generate comprehensive compliance scores by domain
- Prioritize gaps by risk severity and remediation effort
- Track compliance status over time

Knowledge Graph & Regulation Mapping

- Build regulation-policy mapping graphs using NetworkX
- Visualize requirement relationships and dependencies
- Track coverage across compliance domains

Compliance Dashboard

- Develop Gradio dashboard for compliance monitoring
- Provide drill-down views by regulation and domain
- Export audit-ready compliance reports

Key Learning Outcomes

This project equips learners with high-demand AI and RegTech skills, including:

- Large Language Model Integration: Master integration of LLMs using LangChain LCEL for intelligent regulatory analysis and requirement extraction.
- Regulatory Document Engineering: Develop expertise in handling PDF regulations and policy documents with automated parsing.

- Multi-Agent System Development: Gain proficiency in implementing compliance-focused agents with gap detection and scoring.
- Vector Database Implementation: Learn to build and query ChromaDB for semantic regulation-policy matching.
- Knowledge Graph Construction: Master building regulation mapping graphs using NetworkX for coverage visualization.
- Compliance Scoring Systems: Understand techniques for gap severity assessment and remediation prioritization.
- Production-Grade Observability: Learn Langfuse integration for audit trails and compliance logging.
- Clean Architecture & DevOps: Understand software architecture for building maintainable compliance systems.
- Audit Documentation: Generate comprehensive compliance reports and evidence documentation.

Dataset Overview

To build an effective compliance system, ComplianceIQ will analyze 22 regulatory and policy documents covering multiple compliance domains.

Category	Files	Description
Regulatory PDFs	14	IRDAI regulations, MAS guidelines, audit controls
Company Policies	8	Governance, Risk Management, Cyber, Underwriting, Reinsurance
Regulatory Sources	1	Reference list of regulatory sources

Compliance Domains Covered

- Corporate Governance – Board structure, committees, oversight
- Risk Management – Frameworks, assessment, mitigation
- Information Security – Cyber controls, data protection

- Claims Management – Grievance handling, settlement processes
- Reinsurance – Agreement structures, cession practices
- Internal Audit – Controls, assurance, reporting
- Fraud Prevention – Detection, investigation, reporting

Example Queries & Outputs

User Query	System Response
"Analyze compliance with IRDAI corporate governance requirements."	Returns detailed gap analysis with coverage scores and remediation recommendations.
"What are our highest priority compliance gaps?"	Lists gaps sorted by severity with remediation effort estimates.
"Show policy coverage for cyber security regulations."	Displays coverage matrix mapping policies to security requirements.
"Generate audit documentation for risk management."	Produces comprehensive compliance report with evidence references.
"Compare our policies against MAS requirements."	Provides side-by-side comparison with gap identification.
"What evidence do we need for the upcoming IRDAI audit?"	Lists required documentation and evidence by regulatory requirement.
"Track compliance status across all domains."	Displays dashboard view of compliance scores by domain.
"What policy changes are needed for new regulations?"	Identifies policy updates required for regulatory changes.

Technologies Used

Component	Technology
Database	ChromaDB (vector storage and semantic search)
AI Model	OpenAI GPT-4o-mini (regulatory analysis)
Embeddings	OpenAI text-embedding-3-small
Agent Framework	LangChain v1.0+ with LCEL
Knowledge Graph	NetworkX
Data Validation	Pydantic v2
Observability	Langfuse (audit trails, compliance logging)

Frontend	Gradio (compliance dashboard)
Backend Processing	Python (PDF parsing, policy analysis)
Deployment	Local / Google Colab

UI Feasibility

ComplianceIQ follows a dashboard-centric UI approach using Gradio, providing comprehensive visibility into compliance status across all regulatory domains.

Key Features

- Tab-Based Navigation: Separate modules for Regulation Analysis, Gap Detection, Policy Coverage, and Reports

Independent Interfaces

- Compliance Dashboard: Overview of compliance scores across all domains and regulations
- Gap Analysis: Detailed gap listings with severity, remediation steps, and evidence requirements
- Policy Mapping: Visual representation of regulation-policy relationships

User Experience

- Drill-Down Capability: Navigate from summary to detailed gap analysis
- Export Capabilities: Generate audit-ready reports and documentation
- Trend Tracking: Monitor compliance improvements over time

Suggested Implementation Pipeline

This is a suggested pipeline, and learners are encouraged to explore their own approaches based on their expertise and tool preferences.

Week 1: Environment and Data Setup

- Set up development environment (local or Google Colab)
- Configure API keys: OpenAI, Langfuse
- Load regulatory PDFs and policy documents

Week 2: Document Processing

- Parse PDF regulations and DOCX policies
- Extract requirements and policy statements
- Categorize by compliance domain

Week 3: Vector Store and Embeddings

- Set up ChromaDB for regulations and policies
- Create embeddings for semantic matching
- Implement regulation-policy search

Week 4: Single Agent Design

- Build Regulation Monitor Agent with requirement extraction
- Build Gap Detector Agent with severity scoring
- Build Policy Analyzer Agent with coverage assessment

Week 5: Multi-Agent Orchestration

- Implement gap detection workflow
- Build evidence collection mechanisms
- Generate comprehensive compliance scores

Week 6: Knowledge Graphs

- Build regulation-policy mapping graphs with NetworkX
- Map requirement relationships and dependencies
- Visualize coverage across domains

Week 7: Observability and Tracing

- Integrate Langfuse for audit trails
- Implement compliance logging
- Track assessment history

Week 8: Production Deployment

- Full system integration and testing
- Build Gradio compliance dashboard
- Generate assessment reports

Evaluation Criteria

Criterion	Weight	Description
Technical Implementation	40%	Correct use of LCEL, embeddings, agents, knowledge graphs
Compliance Accuracy	20%	Proper requirement extraction, gap identification, scoring
System Architecture	15%	Clean design, proper orchestration, domain coverage

Code Quality	15%	Readable code, documentation, consistent style
Production Readiness	10%	Observability, UI functionality, audit support

Future Enhancements

- Real-time regulatory change monitoring and alerts
- Automated policy update suggestions
- Multi-jurisdiction compliance comparison
- Integration with GRC platforms
- Predictive compliance risk scoring
- Automated evidence collection workflows
- Compliance training content generation
- Regulatory reporting automation

Getting Started

- Basic knowledge of Python & PDF processing
- Set up development environment (local or Google Colab)
- Configure API keys: OpenAI, Langfuse
- Install necessary libraries: LangChain, ChromaDB, PyPDF2, NetworkX, Gradio
- Upload regulatory PDFs and policy documents to designated directories

Reference Material

- LangChain Documentation: https://python.langchain.com/docs/introduction/

- ChromaDB Documentation: https://docs.trychroma.com/
- Langfuse Documentation: https://langfuse.com/docs
- NetworkX Documentation: https://networkx.org/documentation/stable/
- Pydantic Documentation: https://docs.pydantic.dev/latest/
- Gradio Documentation: https://www.gradio.app/docs
- IRDAI Regulations: https://www.irdai.gov.in/
- MAS Guidelines: https://www.mas.gov.sg/

More Use-Cases in Regulatory Compliance

- Automated regulatory filing preparation
- Cross-border compliance harmonization
- Third-party vendor compliance assessment
- Compliance training needs identification
- Regulatory penalty risk assessment